

Heart Disease Prediction Using Logistic Regression

Mini Project Report

Submitted by

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Under the guidance of

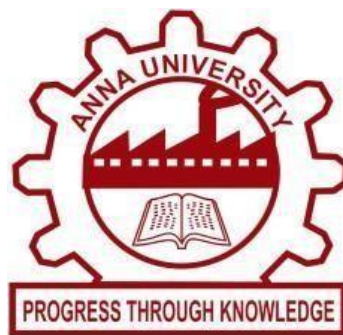
Dr. A. Muthumari

*in partial fulfilment of requirements for
award of Degree of*

BACHELOR OF ENGINEERING

in

Computer Science and Engineering



Anna University Regional Campus, Madurai

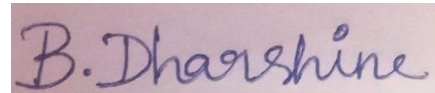
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Declaration

We undersigned, hereby declare that the Mini project titled “**Heart Disease Using Logistic Regression**” submitted in partial fulfilment for the award of Degree of Engineering of Anna University Regional Campus, Madurai is a bonafide record of work done by me under the guidance of **Dr. A. Muthumari** of **Department of Computer Science and Engineering, Anna University Regional Campus, Madurai**. This report has not been previously published for the award of any degree, diploma, or similar title of any University.

Signature



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CERTIFICATE

This is to certify that the report titled **“Heart Disease Prediction Using Logistic Regression”** being submitted **Dharshine B (910023104005)** and **Rishiyanth G (910023104008)** in partial fulfilment of the requirements for the award of the of Degree of Engineering, is a bonafide record of the project work done by **Dharshine B (910023104005)** and **Rishiyanth G (910023104008)** of **Department of Computer Science and Engineering, Anna University Regional Campus, Madurai.**

Dr. A. Muthumari
Assistant Professor

Acknowledgement

Through this acknowledgement we express our sincere gratitude towards all those people who helped me in this project, which has been a learning experience.

This space wouldn't be enough to extend our warm gratitude towards my project guide **Dr. A. Muthumari, Assistant Professor of Department of Computer Science and Engineering, Anna University Regional Campus, Madurai** his efforts in coordinating with my work and guiding in right direction.

It would be injustice to proceed without acknowledging those vital supports we received from our beloved classmates and friends, without whom we would have been half done.

we also use this space to offer our sincere love to my parents and all others who had been there, helping me walk through this work.

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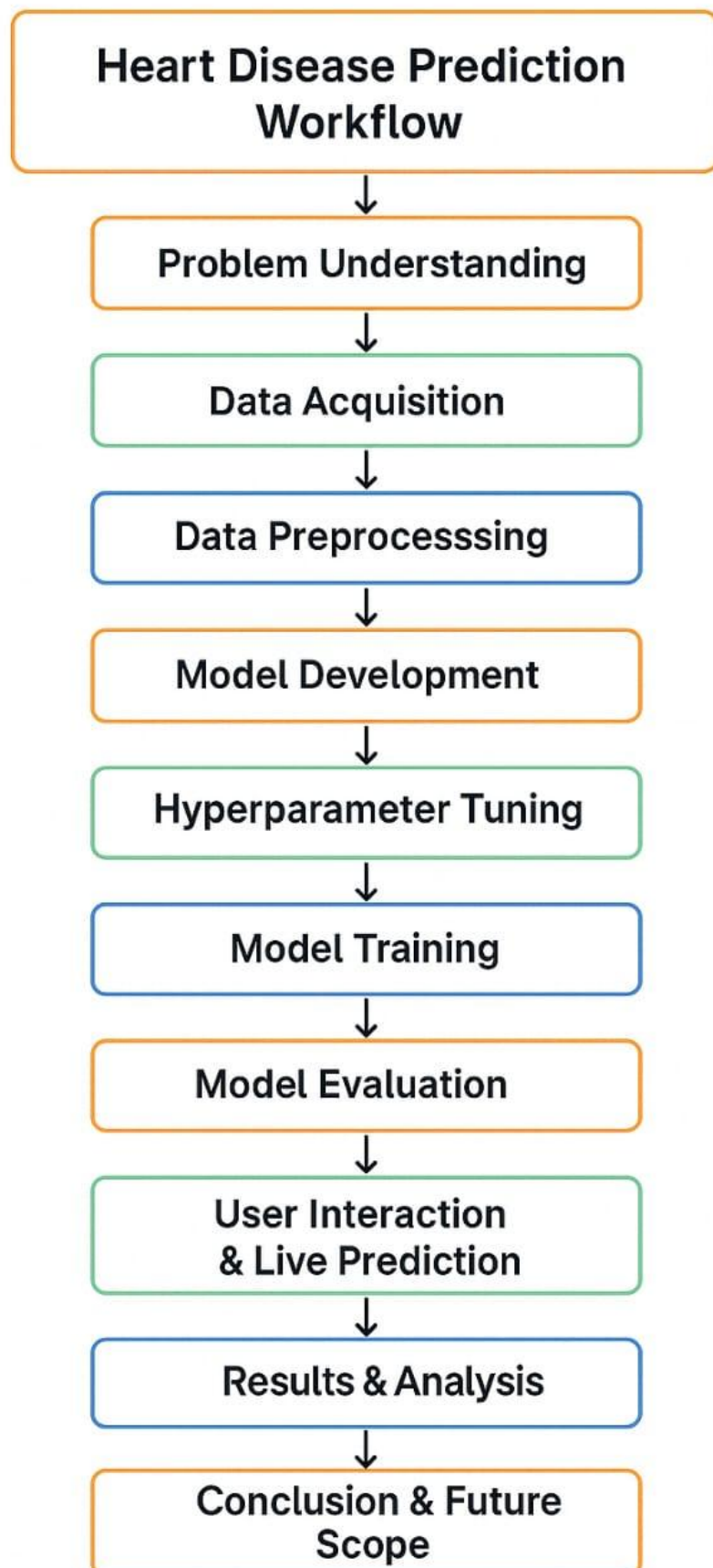
INTRODUCTION

Heart disease is one of the leading causes of death globally, accounting for a large percentage of mortality in both developed and developing nations. Early diagnosis and risk prediction of heart disease is crucial to reducing fatalities and ensuring timely intervention. With the advancement of computational methods, machine learning techniques have become effective tools to aid in medical diagnosis. This project aims to develop a logistic regression-based classifier to predict the presence of heart disease in patients using clinical and demographic data.

PROBLEM STATEMENT

The objective is to build a logistic regression classifier to predict whether a patient has heart disease or not, using the provided dataset. The classifier will take multiple patient attributes (like age, cholesterol level, blood pressure, etc.) and output a binary prediction. The company requires that the model not only provides class labels but also the probabilities associated with each prediction. We will manually implement logistic regression using stochastic gradient descent (SGD), ensuring no use of specialized machine learning libraries like `sklearn.linear_model.LogisticRegression` during model development.

WORK FLOW



Objective

To build a machine learning classifier (using logistic regression) that predicts the presence of heart disease in patients and to compare the performance of our custom implementation with a standard library model. The classifier must output both class labels and prediction probabilities.

Proposed Algorithm

We propose to solve the L1-penalized logistic regression objective:

$$\min_{\{w,b\}} (1/n) \sum \log(1 + \exp(-y_i(w^T x_i + b))) + C \|w\|_1$$

The optimization will be solved via Stochastic Gradient Descent (SGD). Derived gradients:

- Gradient w.r.t weights (w): $-y_i x_i / (1 + \exp(y_i(w^T x_i + b))) + C \cdot \text{sign}(w)$
- Gradient w.r.t bias (b): $-y_i / (1 + \exp(y_i(w^T x_i + b)))$

We will use a decaying learning rate schedule and terminate iterations based on a convergence threshold or a maximum number of epochs.

Algorithm Integration

The classifier will be implemented as a Python class with methods:

- Constructor: Initialize hyperparameters (C, learning rate, etc.)
- Train: Optimize weights and bias using SGD on the training data
- Predict: Output predicted class labels
- Predict_Prob: Output probabilities

Pipeline:

- Train-test split (80%-20%)
- Normalize features
- Perform k-fold cross-validation for tuning
- Evaluate final model on test data

Result Analysis

-Dataset: 1025 samples, 13 features

-Train size: 820 | Test size: 205

Hyperparameter tuning (example):

C=1.0, Learning Rate=0.005 gave best validation accuracy = 84.5%

Final Test Evaluation:

Accuracy: 0.8				
Classification Report:				
	precision	recall	f1-score	support
0	0.85	0.73	0.78	102
1	0.76	0.87	0.81	103
accuracy			0.80	205
macro avg	0.81	0.80	0.80	205
weighted avg	0.81	0.80	0.80	205
Sklern Accuracy: 0.7951219512195122				
Sklern Classification Report:				
	precision	recall	f1-score	support
0	0.85	0.72	0.78	102
1	0.76	0.87	0.81	103
accuracy			0.80	205
macro avg	0.80	0.79	0.79	205
weighted avg	0.80	0.80	0.79	205

Advantages, Disadvantages & Applications of the Algorithm

Advantages:

- Interpretable model
- Handles linearly separable data well
- L1 penalty promotes feature sparsity

Disadvantages:

- Less effective on complex non-linear data
- Sensitive to outliers
- Requires feature scaling

Applications:

- Heart disease risk prediction

- Medical decision support systems
- Feature selection in healthcare data

Conclusion and Future scope

We successfully built a logistic regression classifier from scratch to predict heart disease. The model performed competitively compared to sklearn's implementation.

Future scope includes extending the model to handle non-linearity, advanced feature engineering, and deploying the model in clinical decision support systems.

References

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